

Niklas Rieken

Curriculum Vitae

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in niklas-rieken



Computer Scientist (B.Sc., M.Sc.) with a PhD in Economics and a focus on algorithmic methods, auction theory, and mathematical optimization. I have practical experience with Python, C/C++, and Linux-based workflows for the automation of technical and data-related processes. I am looking for roles in which I can develop data-driven and software-based solutions for complex problems.

Core Skills

Programming	Python (advanced), Bash (proficient), PHP (proficient), C/C++ (intermediate)
Libraries & Frameworks	numpy, scipy, pandas, matplotlib, SAGE, gurobipy, asyncio, requests, BeautifulSoup, Playwright, flask, SQLAlchemy, FastAPI, python-telegram-bot, tkinter
Tools	vim (advanced), LaTeX (+TikZ) (advanced), git (proficient), Gurobi (proficient)
Systems	GNU/Linux (proficient), AWS (basics, EC2, S3)
Web & Data	HTML (proficient), CSS (proficient), SQL (proficient)
Practices	agile Development , DevOps , Clean Code
Methods	Combinatorial Optimization , Graph Algorithms
Languages	German (native), English (fluent), Spanish (basics)

Work Experience

since 08/2026	Business Intelligence Specialist , <i>Dept. 3.3 Organisation & IT</i> , EDG Holding Data Analysis and Data Architecture to optimize reporting and forecasting in Controlling and Material Flow Management using Qlik Sense and Nprinting.
09/2019–08/2025	Doctoral Researcher , <i>Chair of Management Science</i> , RWTH Aachen University Researcher in Combinatorial Optimization and Algorithmic Game Theory. Matchings, matroids, mechanism design, teaching courses in mathematical economics. Responsible for IT and some organisational tasks within the group.
11/2017–04/2019	Research Assistant , <i>Chair of Operations Research</i> , RWTH Aachen University Developer for GCG (<i>generic column generation</i> , MIP solver). Performance tests, extracting statistics, and collecting information on vast instance sets.
04/2014–08/2019	Teaching Assistant , <i>Chair of Computer Science 7 / TCS</i> , RWTH Aachen University TA for undergraduate courses in theoretical computer science. Teaching exercise groups, grading assignments and exams.

Education

09/2019–11/2025	Doctorate (PhD) , <i>Economics</i> , RWTH Aachen University Thesis: Matroid Optimization in Auction Theory Final Grade: summa cum laude (highest distinction)
02/2017–05/2019	Master of Science , <i>Computer Science</i> , RWTH Aachen University Thesis: Network Pricing Games with Atomic Splittable Followers Final Grade: 1.4
10/2012–02/2017	Bachelor of Science , <i>Computer Science</i> , RWTH Aachen University Thesis: Alphabets and Machines as Structures and Their Decision Problems Final Grade: 2.5

Projects & Honors

Commencement, 2026	Borchers' Medal , <i>Award for completing doctorate with distinction</i> by the rector of RWTH Aachen University
Teaching Award, 2024	Prize for outstanding teaching activities , 2024 by the Faculty 8 of Economics, RWTH Aachen University
Podcast, 2021	Algo2Go , <i>topics in computer science and applied mathematics</i> , 2021 Co-hosted German podcast with Prof. Dr. Laura Vargas Koch and Dr. Björn Tauer to explain algorithms to a broad audience. Responsible for content planning, research, and production.

Software (selected)

Telegram Bot	jobscanner Scraper for job listings, which notifies the user via Telegram message if a new role is posted. Written in Python (<code>asyncio</code> , <code>sqlite3</code> , <code>Playwright</code> , <code>python-telegram-bot</code>). https://gitlab.com/ttnick/jobscanner
MIP Solver	GCG A solver for Mixed Integer Programs based on SCIP, using Column Generation. Written in C and C++, I worked on supplementary Python scripts (<code>pandas</code> , <code>numpy</code> , <code>matplotlib</code>) to generate testsets for benchmarking. https://gcg.or.rwth-aachen.de
Data Analysis	Exam-Reporting Processing exam results and producing a report as PDF. Written in $\text{\LaTeX}$ and Python (<code>pandas</code> , <code>numpy</code> , <code>matplotlib</code>).
Identity check	EID-Wiwi Tool to check attendance and identity of students in online examinations via Zoom at Faculty 8 at RWTH Aachen University (winter term 2021/22). Written in PHP, HTML and CSS (<code>bootstrap</code>), my contribution here was mainly in the frontend.

Personal Interests

- table tennis
- guitar
- reading

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Dortmund, August 23, 2026